

Collaborators: Jean Lee

1 Properties of Simple Linear Regression

(a)

$$0 = \sum_{i=1}^n e_i$$

$$0 = \sum_{i=1}^n y_i - \hat{y}_i$$

$$0 = \sum_{i=1}^n y_i - \sum_{i=1}^n \hat{y}_i$$

$$0 = n\bar{y} - \sum_{i=1}^n \hat{y}_i$$

$$0 = n\bar{y} - \sum_{i=1}^n \bar{y} + r\sigma_y \frac{x_i - \bar{x}}{\sigma_x}$$

$$0 = n\bar{y} - \sum_{i=1}^n \bar{y} - \sum_{i=1}^n r\sigma_y \frac{x_i - \bar{x}}{\sigma_x}$$

$$0 = n\bar{y} - n\bar{y} - \sum_{i=1}^n r\sigma_y \frac{x_i - \bar{x}}{\sigma_x}$$

$$0 = - \sum_{i=1}^n r\sigma_y \frac{x_i - \bar{x}}{\sigma_x}$$

$$0 = - \sum_{i=1}^n r\sigma_y \frac{x_i}{\sigma_x} + \sum_{i=1}^n r\sigma_y \frac{\bar{x}}{\sigma_x}$$

$$\sum_{i=1}^n r\sigma_y \frac{x_i}{\sigma_x} = \sum_{i=1}^n r\sigma_y \frac{\bar{x}}{\sigma_x}$$

$$r\sigma_y \frac{n\bar{x}}{\sigma_x} = r\sigma_y \frac{n\bar{x}}{\sigma_x}$$

$$0 = 0$$

(b)

$$\begin{aligned}\hat{\bar{y}} &= \frac{1}{n} \sum_{i=1}^n \hat{y}_i \\ \hat{y}_i &= \frac{1}{n} \sum_{i=1}^n \hat{\theta}_0 + \frac{1}{n} \sum_{i=1}^n \hat{\theta}_1 x_i \\ \hat{\bar{y}} &= \frac{1}{n} \sum_{i=1}^n \bar{y} - \frac{1}{n} \sum_{i=1}^n \hat{\theta}_1 \bar{x} + \frac{1}{n} \sum_{i=1}^n \hat{\theta}_1 x_i \\ \hat{\bar{y}} &= \bar{y} - \hat{\theta}_1 \bar{x} + \hat{\theta}_1 \bar{x} \\ \hat{\bar{y}} &= \bar{y}\end{aligned}$$

(c) Using the last formula given, we can just plug in $\bar{x}$:

$$\begin{aligned}\hat{y} &= \hat{\theta}_0 + \hat{\theta}_1 x \\ \hat{y} &= \bar{y} + r\sigma_y \frac{x - \bar{x}}{\sigma_x}\end{aligned}$$

Plug in x as $\bar{x}$:

$$\begin{aligned}\hat{y} &= \bar{y} + r\sigma_y \frac{\bar{x} - \bar{x}}{\sigma_x} \\ \hat{y} &= \bar{y} + 0 \\ \hat{y} &= \bar{y}\end{aligned}$$

Thus, $(\bar{x}, \bar{y})$ lies on the simple linear regression line.

2 Geometric Perspective of Least Squares

- (a) From the concepts from lecture, recall that for an optimal parameter $\hat{\theta}$, the residual vector e becomes orthogonal to the $\text{span}(\mathbb{X})$. So we know that one vector from the $\text{span}(\mathbb{X})$ will cause the summation of residuals to be zero. So for such a vector $\vec{x} \in \mathbb{X}$, $\vec{e} \cdot \vec{x} = 0$. In this case, we will conveniently prove it with the bias vector:

$$\vec{e} \cdot \vec{1} = 0$$

$$\sum_{i=1}^n e_i = 0$$

- (b) If we considered the data vector $\vec{x}$ and residual vector $\vec{e}$ to be orthogonal, then it geometrically makes sense. Adding vectors x and e will lead to the projection of another vector on to a hyperplane, which happens to be the hyperplane of the regression formula. Thus, the orthogonal projection of the residual vector on $\vec{x}$ has to be true as it minimizes the distance or residual. We've essentially created a triangle with an adjacent side and hypotenuse, and the only way to have the opposite side as minimum as possible is to have it perpendicular to the adjacent side. Connecting it with other parts, minimizing the residual also makes sense in terms of minimizing the loss function. In terms of linear algebra, this is equivalent to $\vec{e}$ being orthogonal to the $\text{span}(\vec{1}, \vec{x})$ and $\forall \vec{x} \in \text{span}(\vec{1}, \vec{x})$. Thus, $\vec{e}$ is also orthogonal to $\vec{x}$!

- (c) Recall that:

$$\hat{\mathbf{Y}} = \mathbb{X}\hat{\theta}$$

$$\hat{\mathbf{Y}} = \vec{1}\hat{\theta} + \vec{x}\hat{\theta}$$

$\vec{x}$ is just generalized as the design matrix without the bias vector. Since $\hat{\mathbf{Y}}$ is just a linear combination of $\mathbb{X}$, we can conclude that $\hat{\mathbf{Y}} \in \text{span}(\mathbb{X})$. Thus, using answers from the previous problem that: $\text{span}(\vec{1}, \vec{x})$ and $\forall \vec{x} \in \text{span}(\vec{1}, \vec{x})$. Thus, $\vec{e}$ is also orthogonal to $\vec{x}$, we can also deduce that $\hat{\mathbf{Y}}$ is also orthogonal to the residual vector!

3 Properties of a Linear Model With No Constant Term

$$R(\gamma) = \frac{1}{n} \sum_{i=1}^n (y_i - \gamma x_i)^2$$

$$R(\gamma) = \frac{1}{n} \sum_{i=1}^n y_i^2 - 2\gamma x_i y_i + \gamma^2 x_i^2$$

$$R(\gamma) = \frac{1}{n} \sum_{i=1}^n y_i^2 - \frac{1}{n} \sum_{i=1}^n 2\gamma x_i y_i + \frac{1}{n} \sum_{i=1}^n \gamma^2 x_i^2$$

$$R(\gamma) = \frac{1}{n} \sum_{i=1}^n y_i^2 - \frac{2\gamma}{n} \sum_{i=1}^n x_i y_i + \frac{\gamma^2}{n} \sum_{i=1}^n x_i^2$$

$$R(\gamma) \frac{d}{d\gamma} = -\frac{2}{n} \sum_{i=1}^n x_i y_i + \frac{2\gamma}{n} \sum_{i=1}^n x_i^2$$

$$0 = -\frac{2}{n} \sum_{i=1}^n x_i y_i + \frac{2\gamma}{n} \sum_{i=1}^n x_i^2$$

$$\frac{2}{n} \sum_{i=1}^n x_i y_i = \frac{2\gamma}{n} \sum_{i=1}^n x_i^2$$

$$\hat{\gamma} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

We can also prove that we are minimizing on a convex function by taking the second derivative.

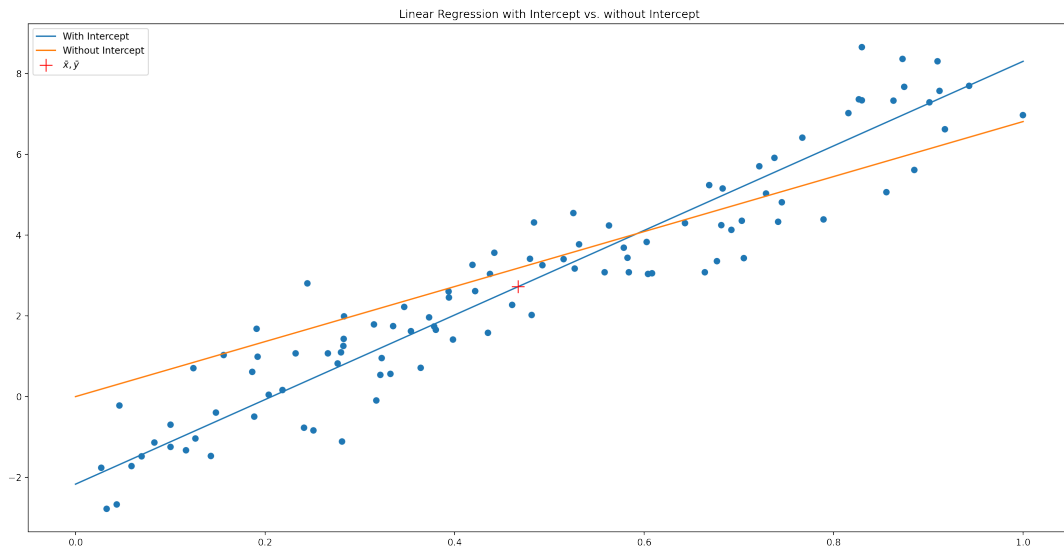
$$R(\gamma) \frac{d^2}{d\gamma^2} = \frac{2}{n} \sum_{i=1}^n x_i^2$$

$$0 \leq \frac{2}{n} \sum_{i=1}^n x_i^2$$

Since x_i is being squared and we are taking the sum, we are guaranteed a nonnegative number, thus we are also minimizing over the function.

4 Properties of a Linear Model With No Constant Term

- (a) False. Without the bias term, we cannot confidently prove for all cases of $\vec{x}$ that $\vec{1}$ is in the $\text{span}(x)$, as having the bias vector $\vec{1}$ was the whole point being able to prove that $\sum e_i = 0$ in the previous parts.
- (b) True. In general, we have proven that $\vec{e}$ is orthogonal to $\text{span}(\vec{x})$ and $\vec{x} \in \text{span}(\vec{x})$ for any $\vec{x}$, as this is geometrically the line/hyperplane of best fit, where we have to use the orthogonal projections as the residual vectors.
- (c) True. Again, from the previous answers, we have proven that the general case of $\hat{Y}$ should be orthogonal to $\vec{e}$, as $\hat{Y}$ is only a linear combination (or rather scalar) in $\hat{Y} = \hat{\theta}\vec{x}$, thus, $\hat{Y}$ should remain in the $\text{span}(\vec{x})$ as $\vec{e}$ is still orthogonal to $\hat{Y}$ and $\text{span}(\vec{x})$.
- (d) False. Geometrically, this fails as the hyperplane/line of best fit would be anchored at the origin without an intercept. Mathematically, recall in a previous question that we needed the bias term to prove that $(\bar{x}, \bar{y})$ was on the regression line, or else it is $\hat{Y} = \hat{\theta}\bar{x}$ without the intercept term.



- (e)
- We see that the simple linear regression model without an intercept term in fact does not have the same slope as the model with an intercept term. (Two lines intersect, thus they are not parallel or have same slope).
 - One shortcoming of simple linear regression model without an intercept term is that we overshoot negative values and undershoot positive values. This imbalance causes inaccurate predictions.

5 MSE “Minimizer”

- (a) The MSE loss function can be viewed as a sum of n quadratic terms, each of which can be treated as a function of x_i

- (b) g_i is convex if it's second derivative, $\frac{d^2 g_i(x)}{d\gamma^2}$:

$$\frac{dg_i(x)}{d\gamma} = \frac{1}{n}(-x_i)(y_i - \gamma x_i)$$

$$\frac{dg_i(x)}{d\gamma} = -\frac{1}{n}x_i y_i + \frac{1}{n}\gamma x_i^2$$

$$\frac{d^2 g_i(x)}{d\gamma^2} = \frac{1}{n}x_i^2$$

$$\frac{1}{n}x_i^2 \geq 0$$

We know that squaring x_i will give a value that is greater than 0, thus the equation $g(\gamma)$ is convex.

- (c) From calculus, we learned that we can find slope of an equation at 0 by taking the first derivative and setting it to 0. Then we can also determine the convexity of the equation by taking the derivative again and figuring out if the function is concave or convex if function is greater or less than 0. Intuitively, the first derivative encodes the slope of the equation at every point for $g_i(x)$. If we are given the condition of convexity, then that means a slope of 0 at a critical point is a minimum or local minima.

- (d) i We can just prove $g(x) + h(x)$ is convex by plugging it in.

$$(g + h)(cx_1 + (1 - c)x_2) \leq cg(x_1) + (1 - c)g(x_2) + ch(x_1) + (1 - c)h(x_2)$$

$$(g + h)(cx_1 + (1 - c)x_2) \leq c(g(x_1) + h(x_1)) + (1 - c)(g(x_2) + h(x_2))$$

$$(g + h)(cx_1 + (1 - c)x_2) \leq c(g + h)(x_1) + (1 - c)(g + h)(x_2)$$

Thus, $g(x) + h(x)$ follows the same format of convexity given in the problem, so that must mean taking the sum of convex function still leads to a convex function.

- ii We can rigorously prove by proof by induction for a general case of $n > 2$, where we just take iterative sums and using the previous answer as a base case, but intuitively we know that adding two convex functions should lead to another convex function, as adding functions only really adds up the points in $g(x_i) + h(x_i)$. It also preserves the shape of the function if they are of the same type, e.g. $x^2 + x^4$ is still convex.

(e) We're guaranteed to minimize the MSE loss, as we are squaring terms to remove the negativity.

$$\frac{1}{n} \sum_{i=1}^n (y_i - \gamma x_i)^2$$

$$\frac{1}{n} \sum_{i=1}^n y_i^2 - 2\gamma x_i y_i + \gamma^2 x_i^2$$

Then taking the gradient of either γ, x_i, y_i will lead to a function greater than or equal to 0, proving convexity.